CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 5A: Inference for Single Proportion

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Outline

1. Test for normality
2. CI for proportion
3. Hypothesis testing
4. Using p-values
5. Outlook

Module objectives

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- Contigency tables

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Question

A simple random sample of 826 payday loan borrowers was surveyed to better understand their interests around regulation and costs. 70% of the responses supported new regulations on payday lenders. Is it reasonable to model $\hat{p} = 0.70$ using a normal distribution?

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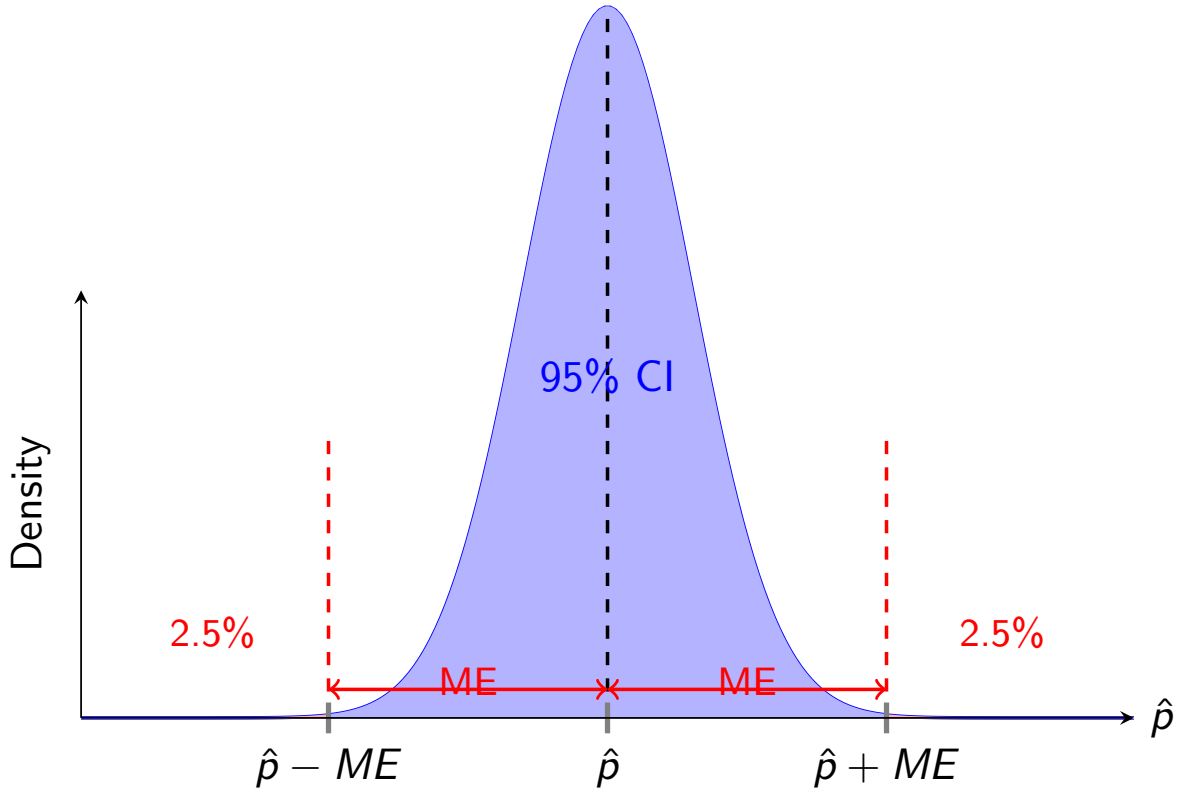
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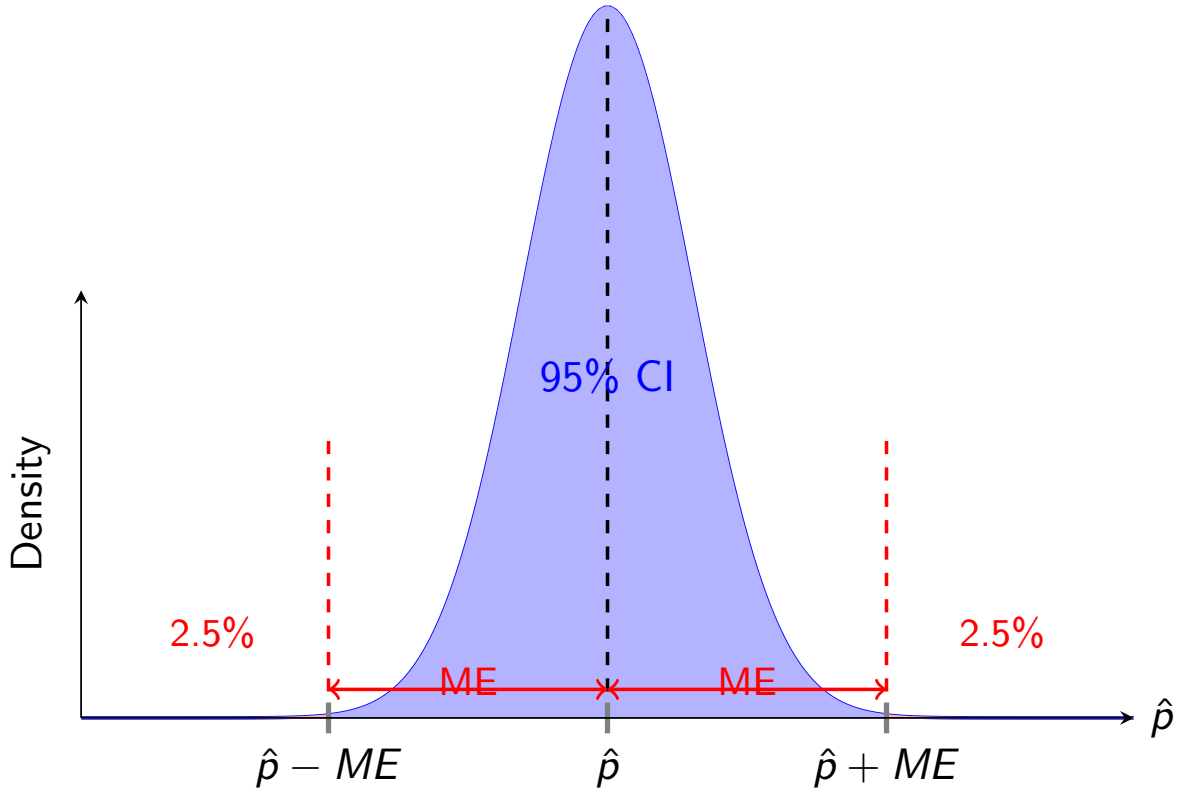
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Note that p is unknown, so we use $\hat{p}$ in its place when computing the standard error.

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- The confidence interval is symmetric around $\hat{p}$
- Margin of error (ME) extends equally in both directions
- CI $= [\hat{p} - ME, \hat{p} + ME]$

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(e) Construct a 95% confidence interval for p, the proportion of adults

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(a) Success-failure condition:

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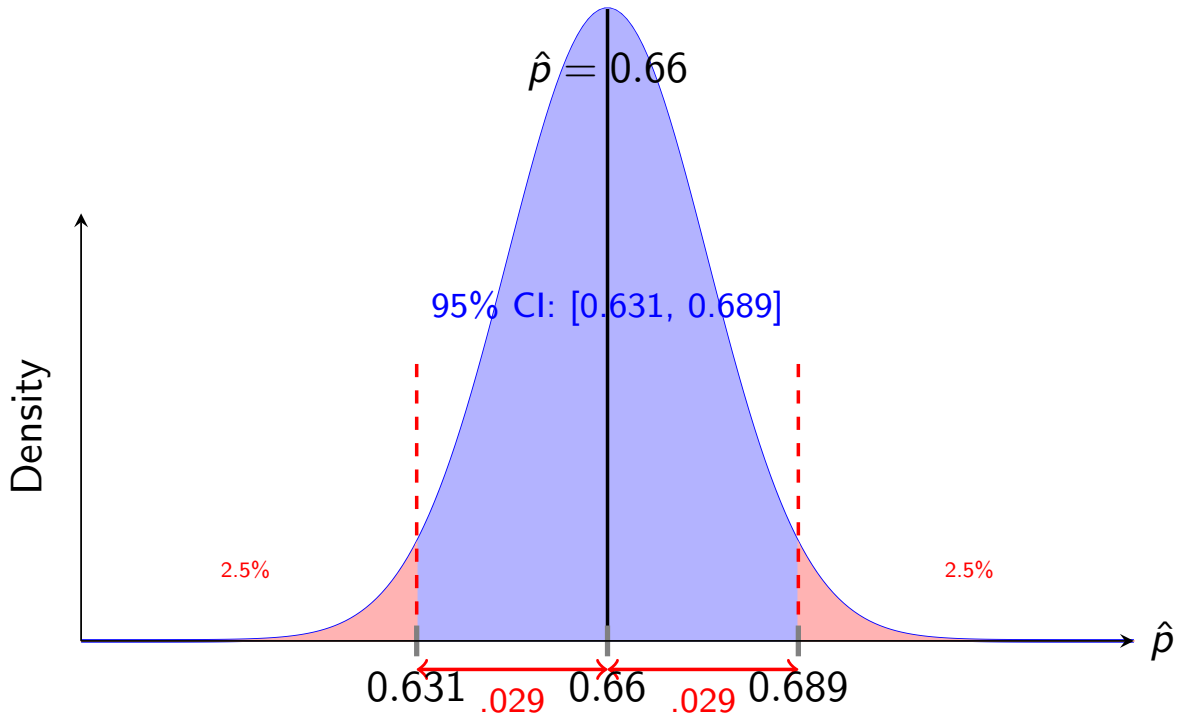
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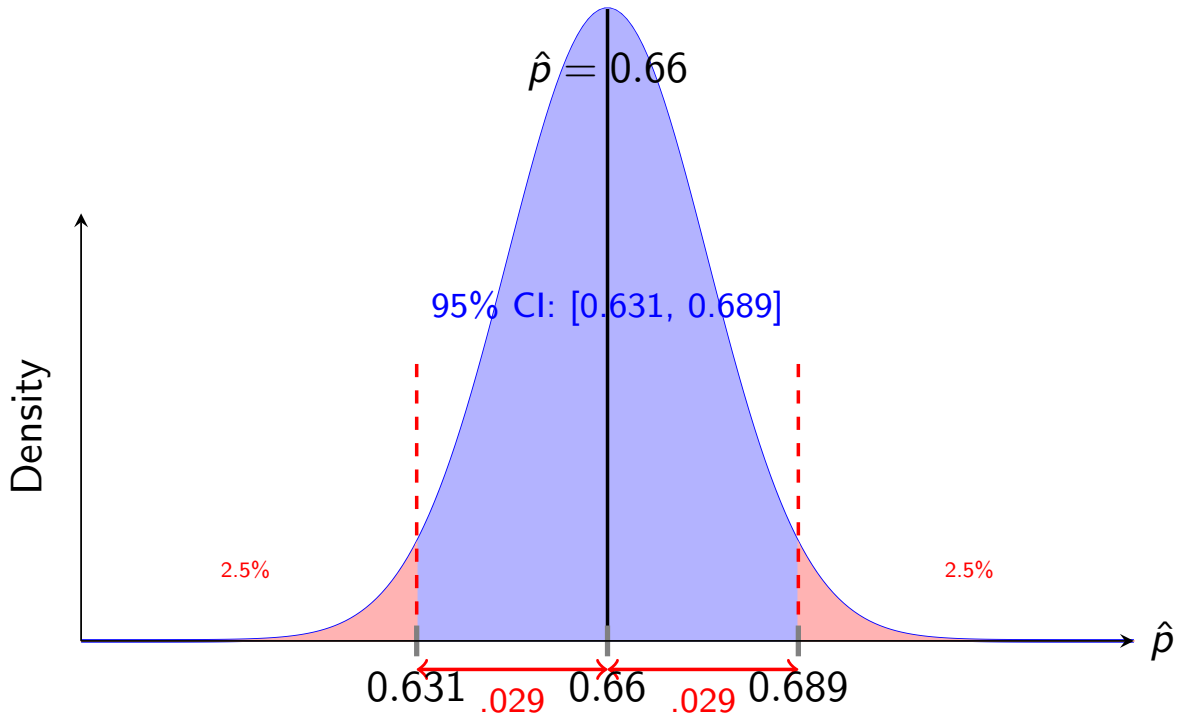
(e) Confidence interval:

$$CI = [\hat{p} - ME, \hat{p} + ME] = [0.66 - 0.0292, 0.66 + 0.0292] = [0.6308, 0.6892]$$

Example 1 (cont.)



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Conclusion: We are 95% confident that between 63.1% and 68.9% of American adults think licensed drivers should be required to retake their road test once they reach 65 years of age.

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6. *Decide.* If the test statistic is in the critical region (i.e. $|z| > z^*$), reject H_0. If not, do not reject H_0 (fail to reject it)

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Thus, to compute the probability (area under curve) of the test statistic, we use the z-score:

$$z = \frac{\hat{p} - p_0}{SE_p} \tag{7}$$

which is normally distributed.

Two-sided tests

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Case A: both tails

Two-sided tests

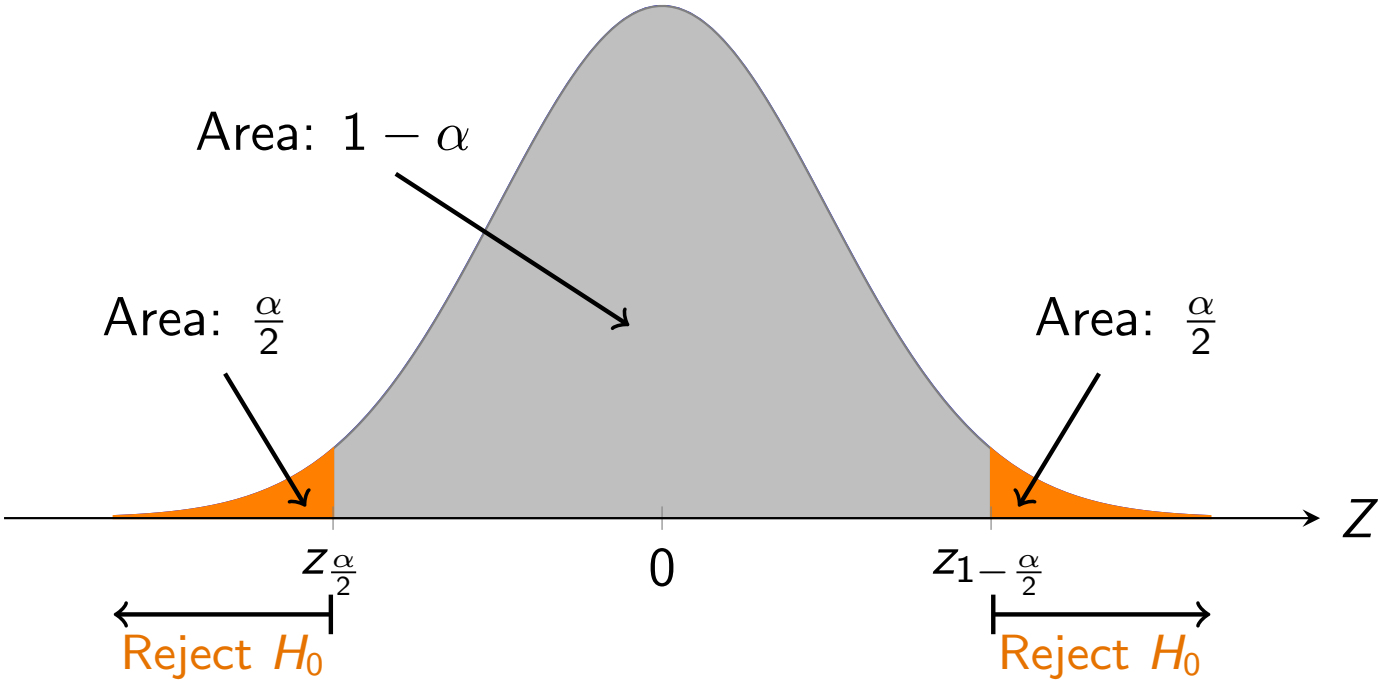
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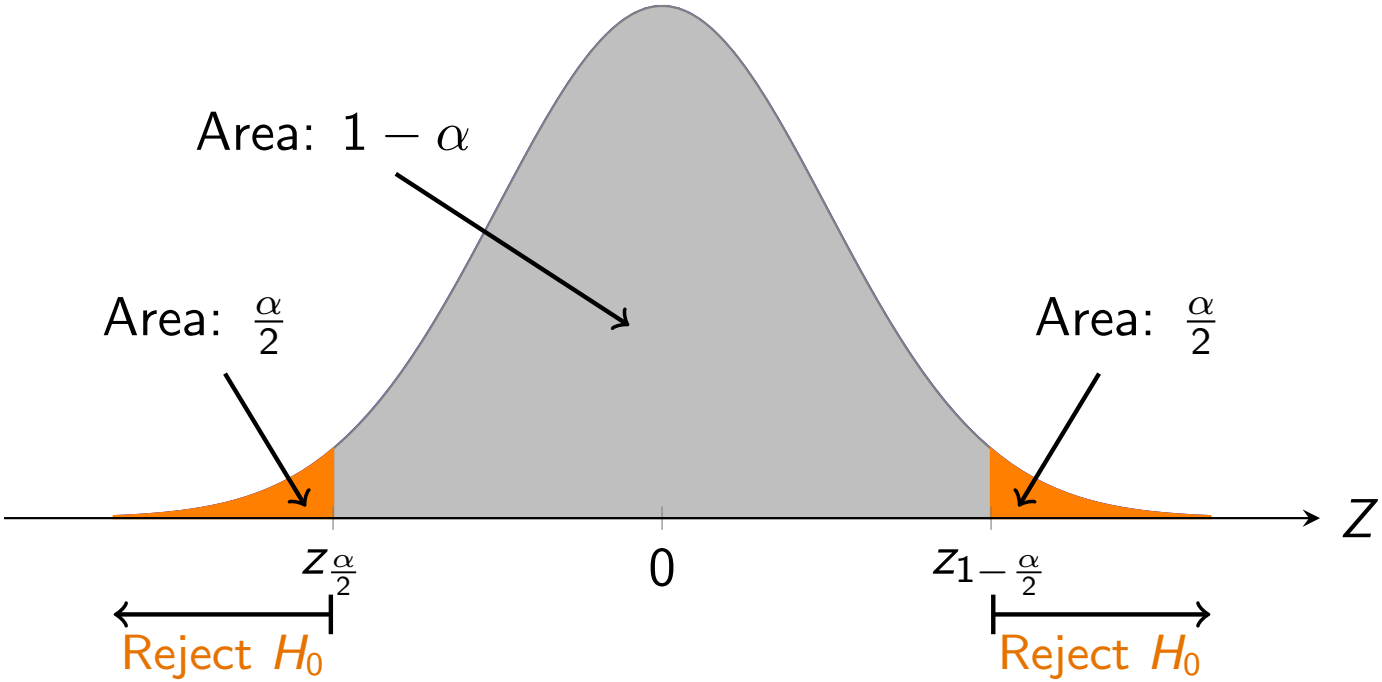
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$$z^* = z_{1-\frac{\alpha}{2}} = \left| z_{\frac{\alpha}{2}} \right| \tag{8}$$

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Case B: upper tail

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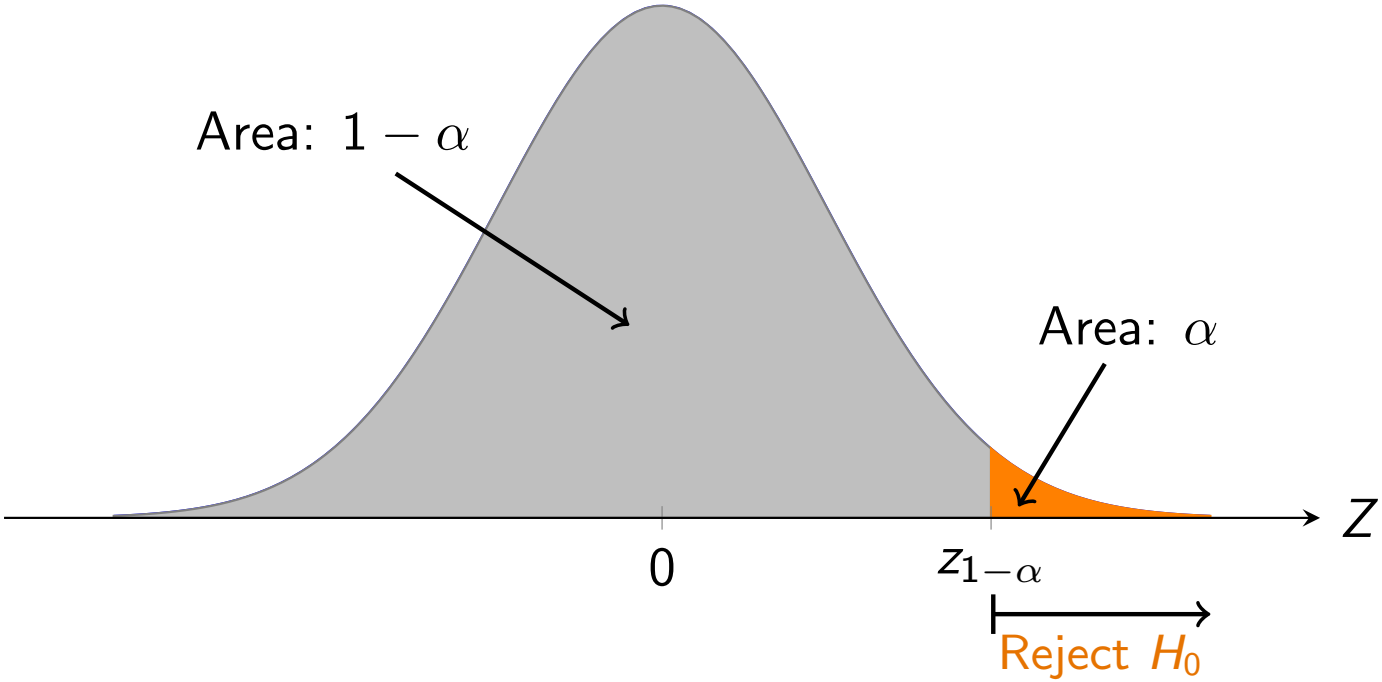
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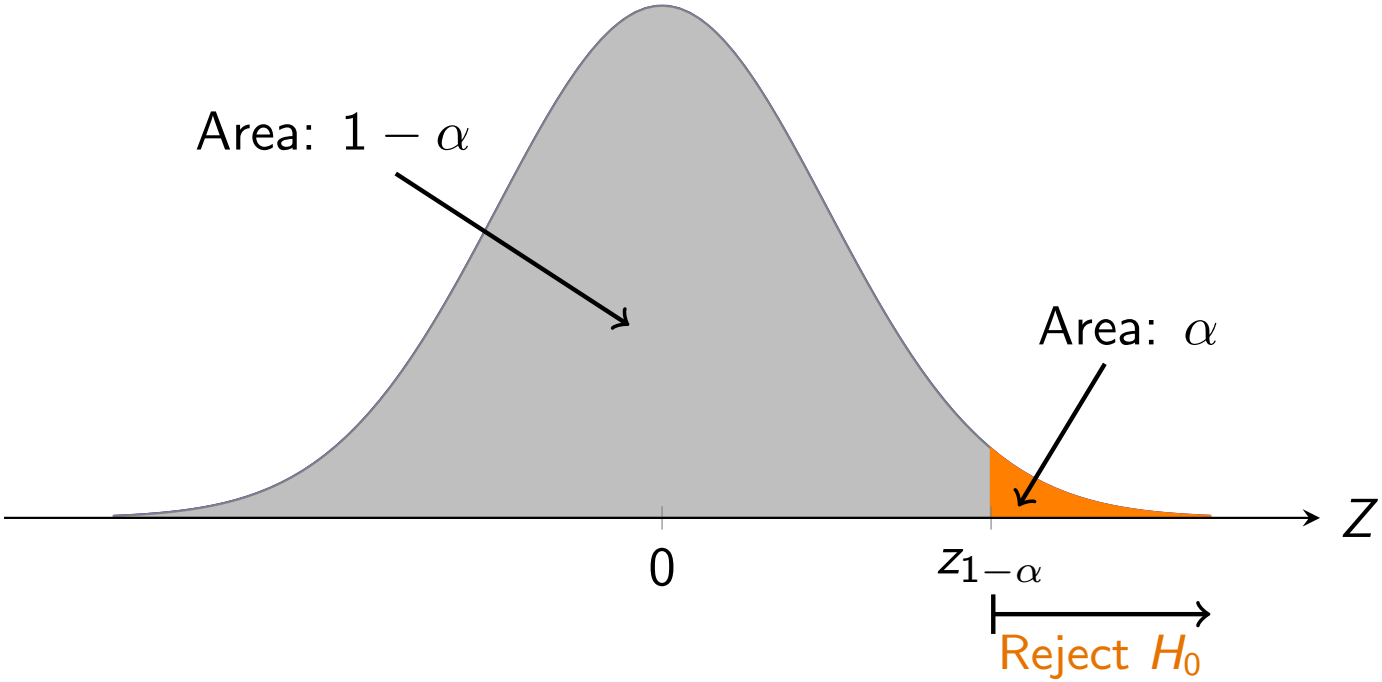
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$$z^* = z_{1-\alpha} 2 \tag{9}$$

One-sided tests (cont.)

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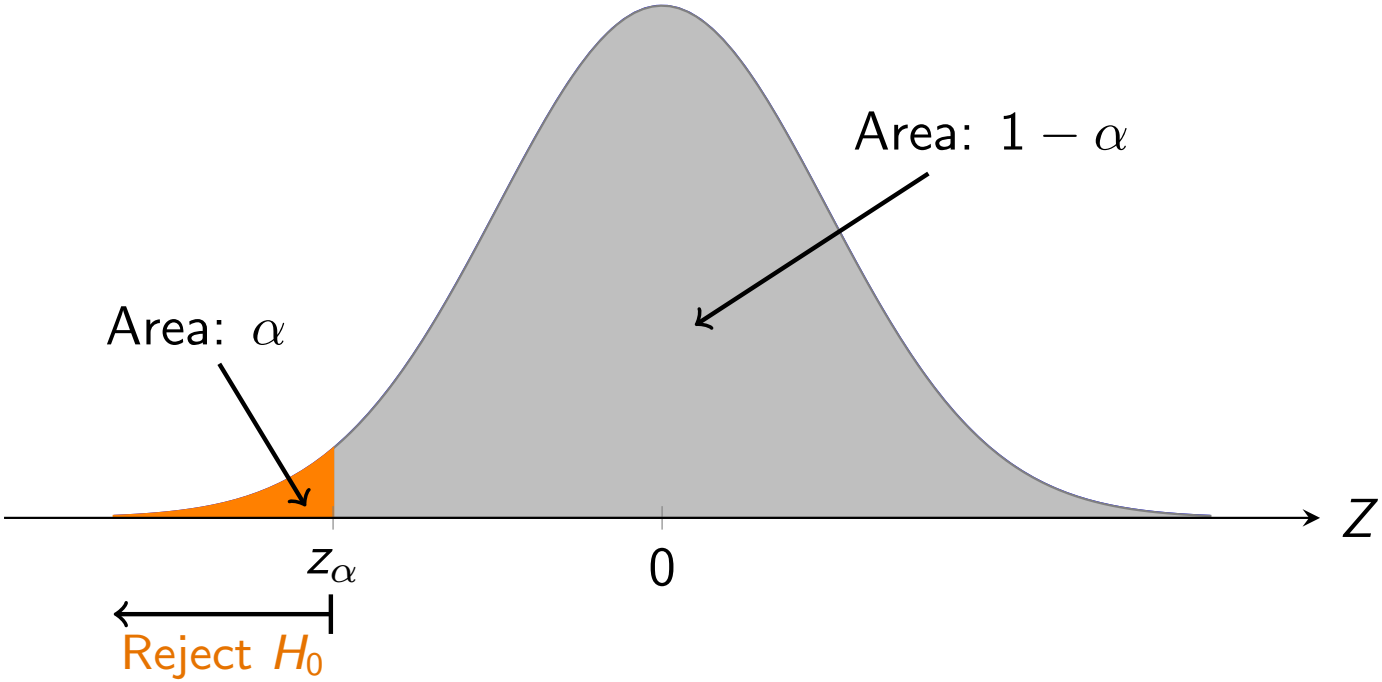
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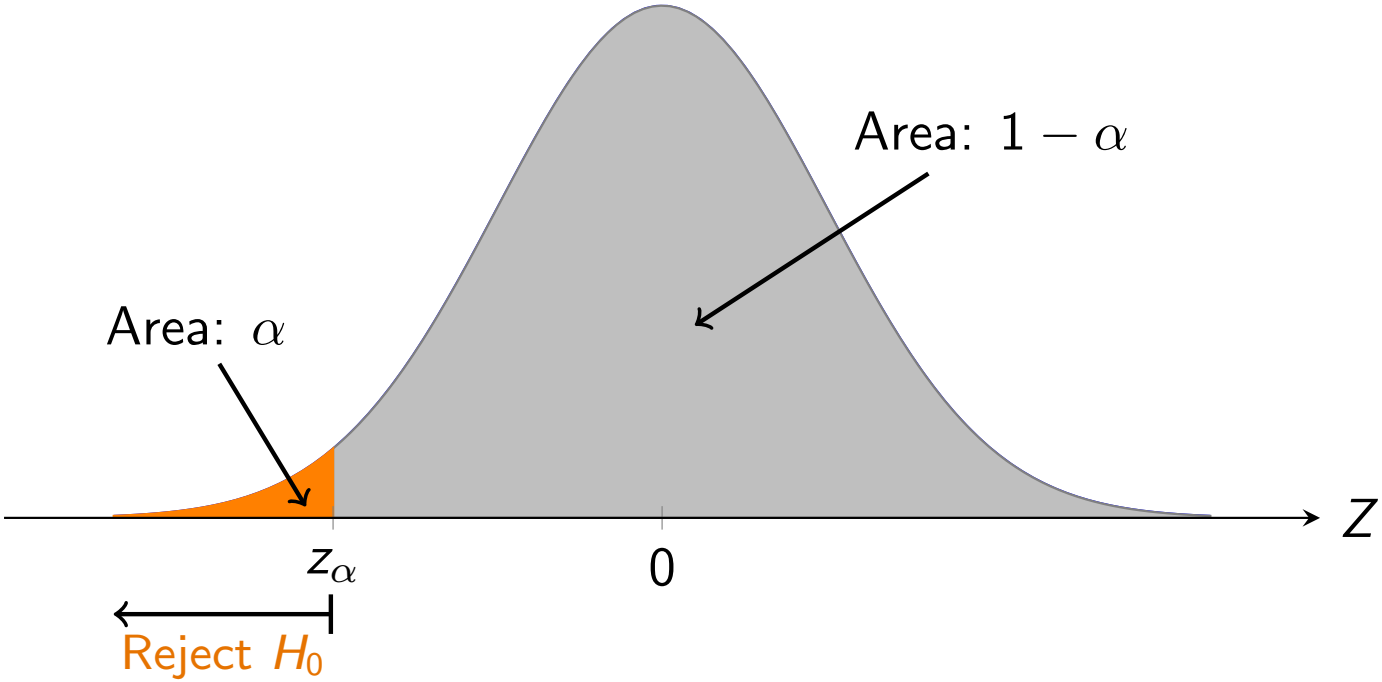
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$$\hat{p} = \frac{289}{400} = .723 \tag{11}$$

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Step 7. Compare test statistic to critical value:

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$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Compute critical value at $\alpha = 0.01$:

$$z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2) = \Phi^{-1}(0.995) = 2.576 \quad \texttt{norm.ppf(0.995)}$$

Step 7. Compare test statistic to critical value:

$$|z| = 8.92 > 2.576 = z_{\alpha/2}$$

Example 2: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Compute critical value at $\alpha = 0.01$:

$$z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2) = \Phi^{-1}(0.995) = 2.576 \quad \texttt{norm.ppf(0.995)}$$

Step 7. Compare test statistic to critical value:

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Step 8. Conclude:

Example 2: Getting enough sleep (cont.)

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$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Compute critical value at $\alpha = 0.01$:

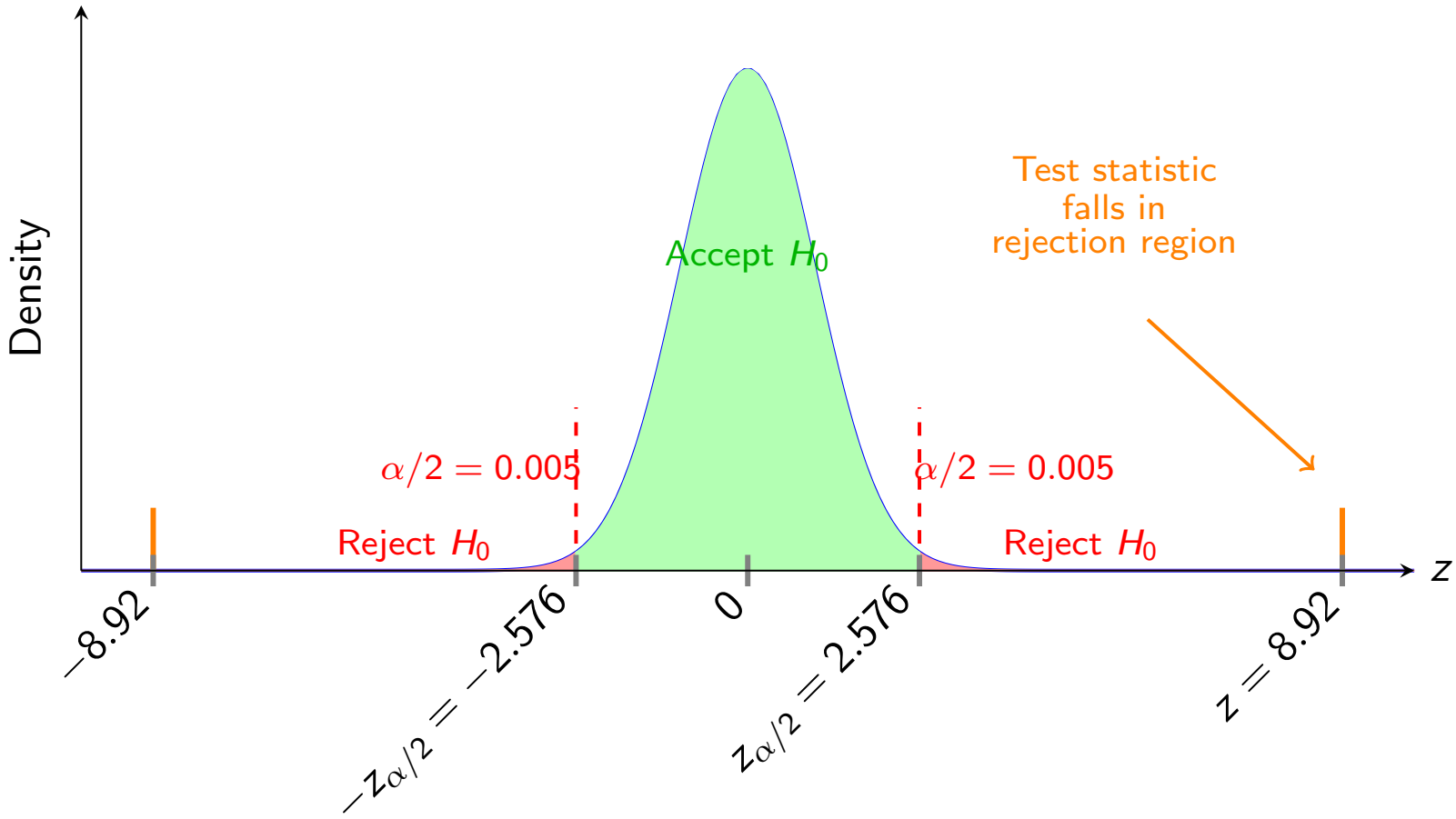
$$z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2) = \Phi^{-1}(0.995) = 2.576 \quad \texttt{norm.ppf(0.995)}$$

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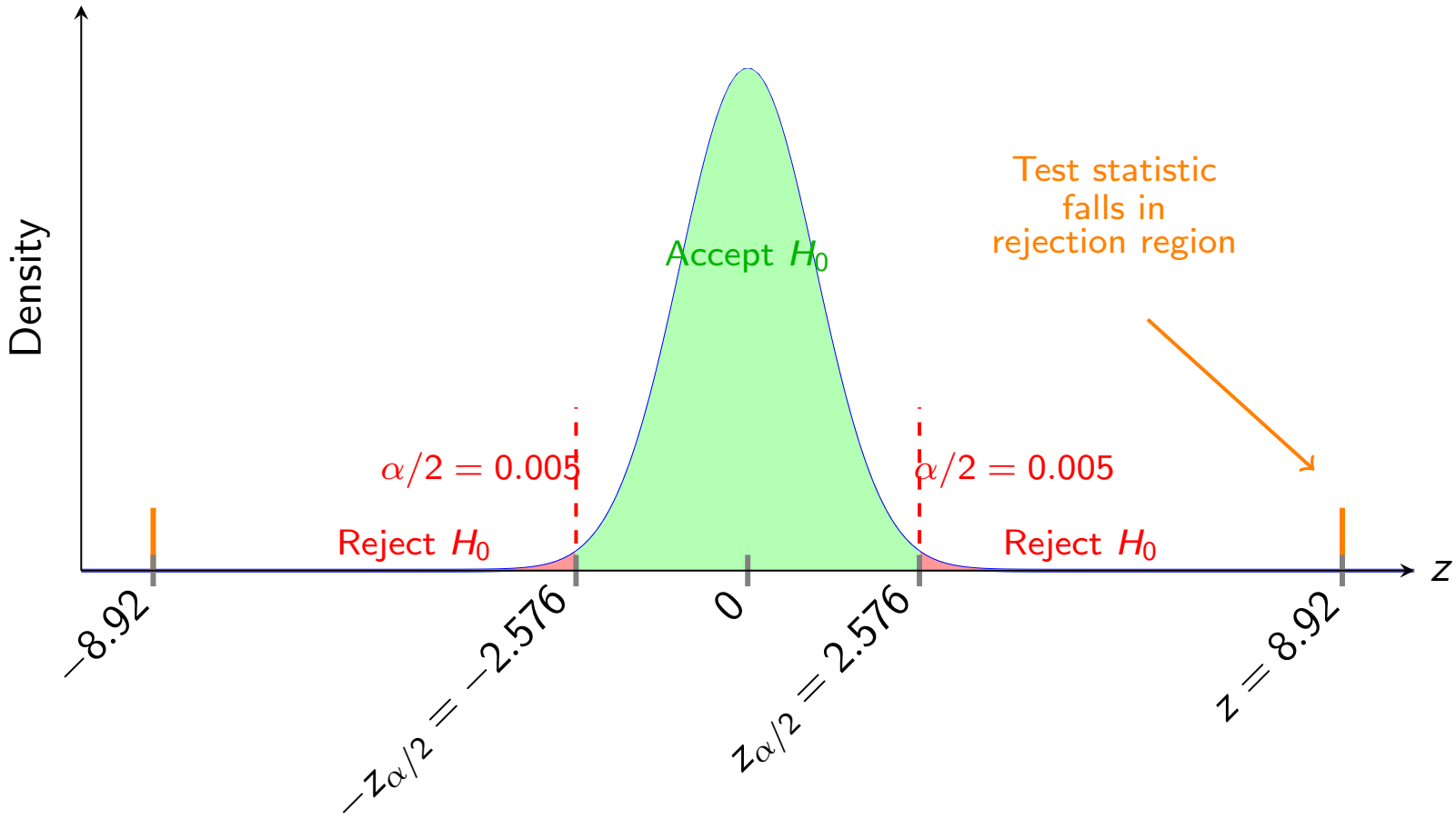
$$|z| = 8.92 > 2.576 = z_{\alpha/2}$$

Step 8. Conclude:
Using a significance level of 0.01, we reject H_0 since $8.92 > 2.576$. Thus, at the 1% significance level, there is sufficient evidence to conclude that true proportion differs from the target value of 0.5.

Example 2: Getting enough sleep (cont.)



Example 2: Getting enough sleep (cont.)



Decision: Since $|z| = 8.92 > 2.576 = z_{\alpha/2}$, we **reject** H_0 at $\alpha = 0.01$ significance level.

Example 3: Equipment quality

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A car manufacturer is considering switching to a new, higher quality piece of equipment that constructs vehicle door hinges. They figure that they will save money in the long run if this new machine produces hinges that have flaws no more than 0.2% of the time. A random sample of 800 hinges produced by the new machine shows that 4 are flawed. At the 0.05 significance level, is there enough evidence to conclude that the new machine produces hinges with a flaw rate greater than 0.2%?

Example 3 (cont.)

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This is a one-sided test (upper-tailed)

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Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Example 3 (cont.)

This is a one-sided test (upper-tailed)

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

$$\hat{p} = \frac{4}{800} = 0.005$$

Example 3 (cont.)

This is a one-sided test (upper-tailed)

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

$$\hat{p} = \frac{4}{800} = 0.005$$

$$SE = \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016$$

Example 3 (cont.)

This is a one-sided test (upper-tailed)

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

$$\begin{aligned} \hat{p} &= \frac{4}{800} = 0.005 \\ SE &= \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016 \\ z &= \frac{\hat{p} - 0.002}{SE} \approx 1.875 \end{aligned}$$

Example 3 (cont.)

This is a one-sided test (upper-tailed)

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

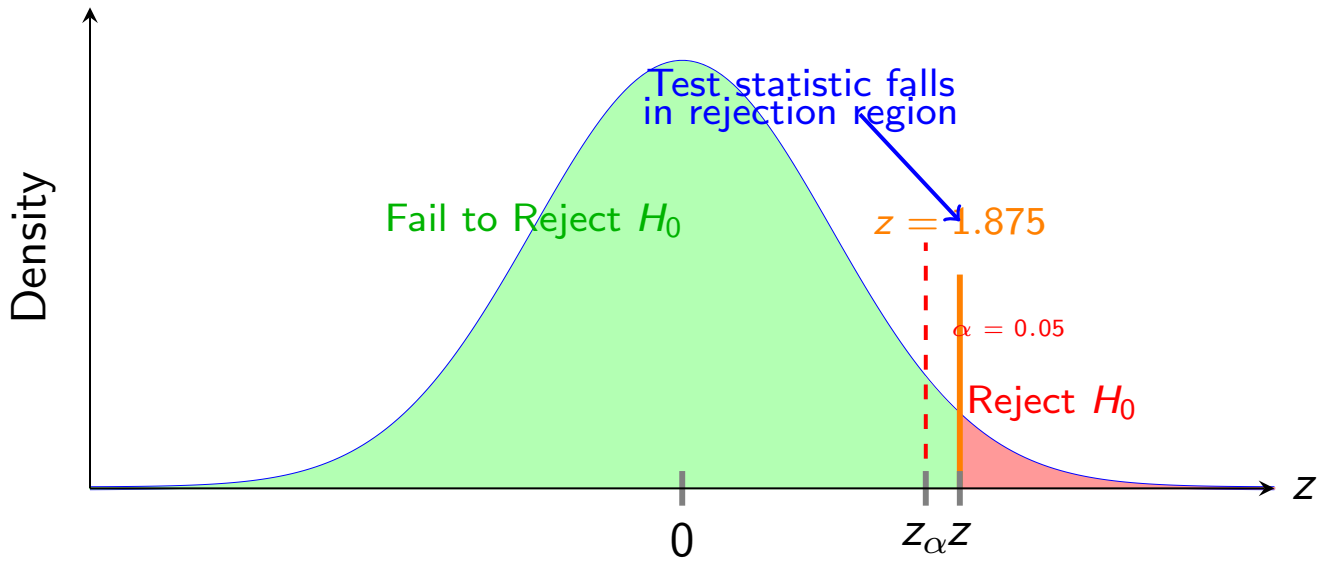
$$\begin{aligned} \hat{p} &= \frac{4}{800} = 0.005 \\ SE &= \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016 \\ z &= \frac{\hat{p} - 0.002}{SE} \approx 1.875 \end{aligned}$$

Critical Value: For $\alpha = 0.05$, the critical value is:

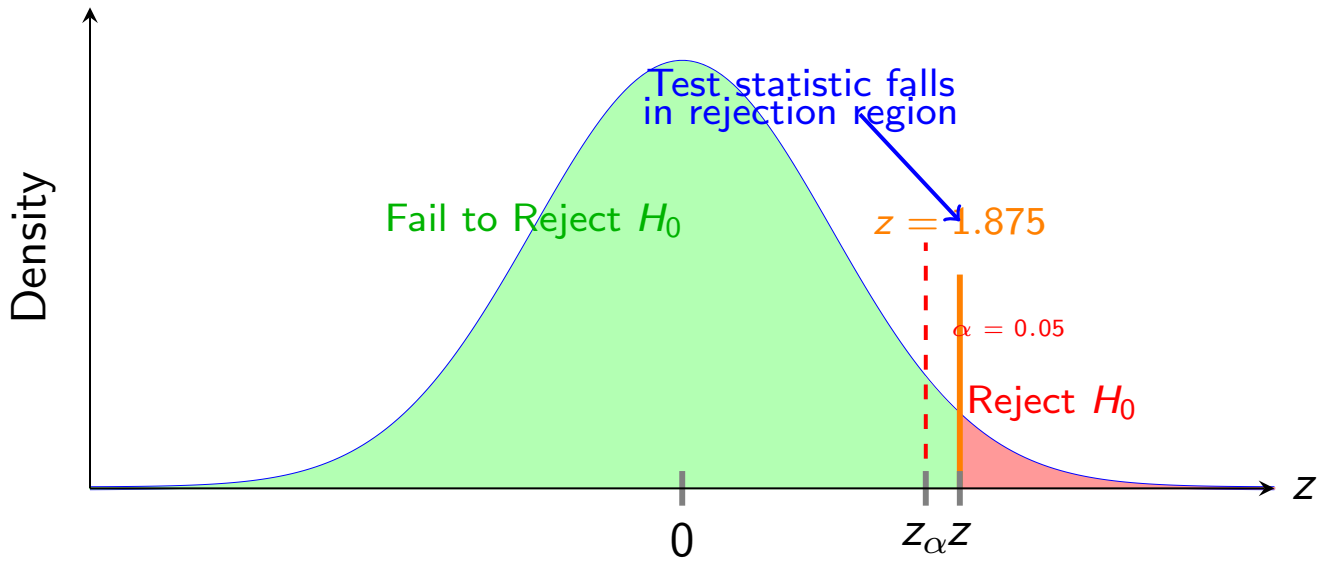
$$z_\alpha = \Phi^{-1}(1-\alpha) = \Phi^{-1}(0.95) = 1.645 \quad \texttt{norm.ppf(0.95)}$$

Example 3 (cont.)

Example 3 (cont.)



Example 3 (cont.)



Decision: Since $z = 1.875 > 1.645 = z_\alpha$, we **reject** H_0 at $\alpha = 0.05$ significance level.

p-value

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The p-value is the probability of obtaining a test statistic value at least as contradictory to H_0 as the value that actually resulted.

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Usefulness of p-value

- Provides more information about the strength of a test

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- Indicates the smallest level at which the data is significant

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Usefulness of p-value

- Provides more information about the strength of a test
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- Can be compared with α irrespective of which type of test was used

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Usefulness of p-value

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- Indicates the smallest level at which the data is significant
- Can be compared with α irrespective of which type of test was used

p-value

The *p*-value is the probability of obtaining a test statistic value at least as contradictory to H_0 as the value that actually resulted. The smaller the *p*-value, the more contradictory are the data to H_0.

Usefulness of p-value

- Provides more information about the strength of a test
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Step 1. Formulate your hypotheses

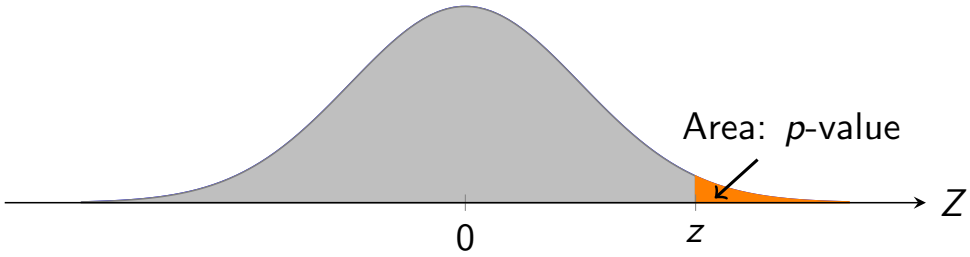
Step 2. Determine the *p*-value from the test statistic

Step 3. Conclude the test based on a chosen level of significance:

1. *p*-value $\leq \alpha \implies$ reject H_0 at level α.
2. *p*-value $> \alpha \implies$ do not reject H_0 at level α.

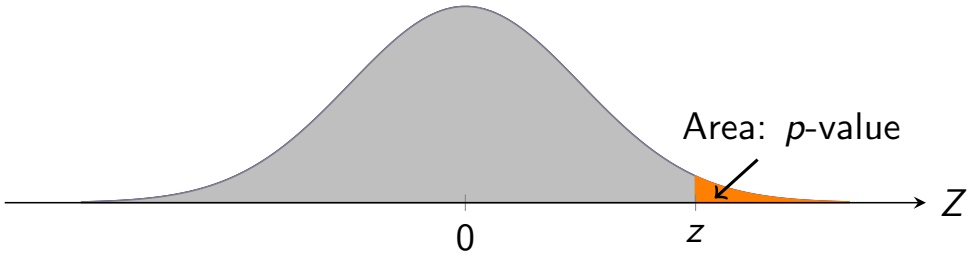
p-value for z tests

p-value for z tests



p-value: area in upper tail

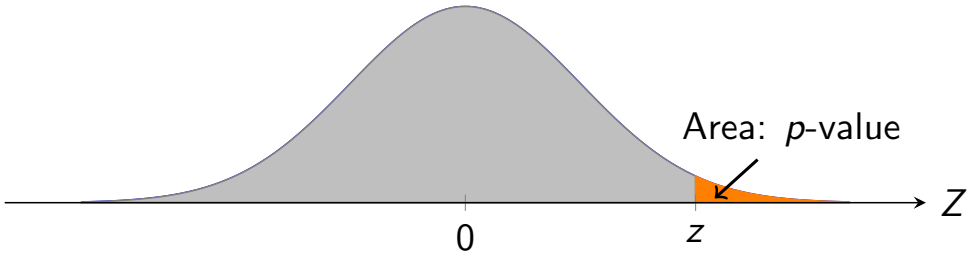
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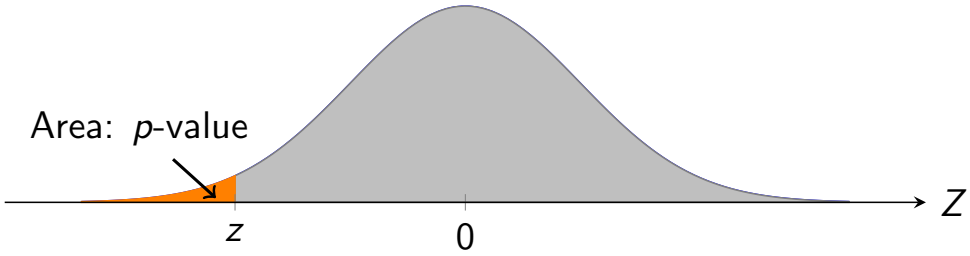


p-value: area in upper tail

$$p = 1 - \Phi(z) \qquad (12)$$

p-value for z tests



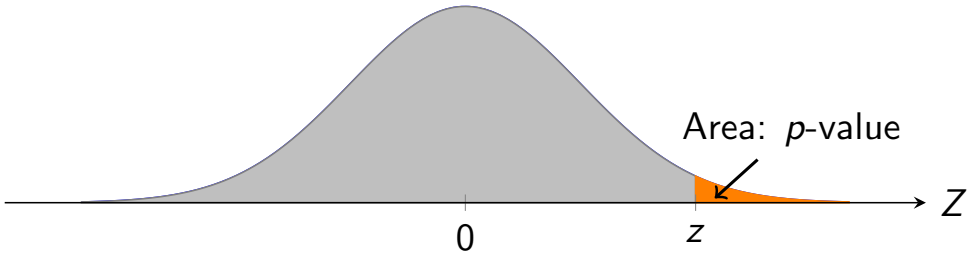


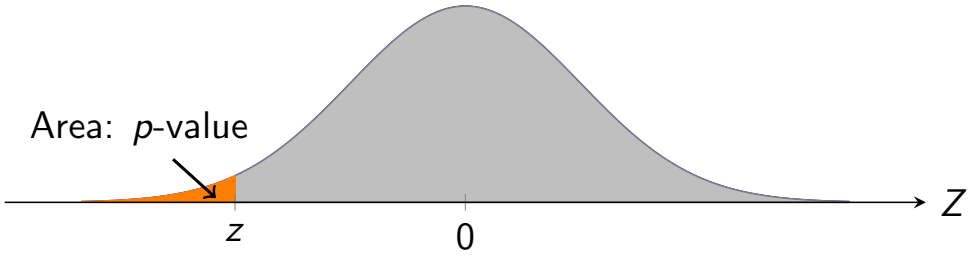
p-value: area in upper tail

$$p = 1 - \Phi(z) \qquad (12)$$

p-value: area in lower tail

p-value for z tests





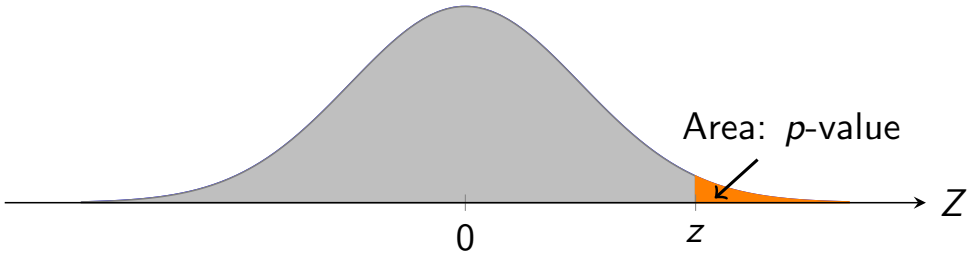
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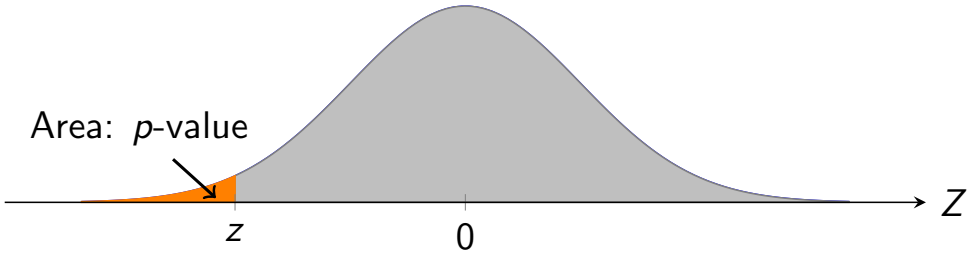
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p-value for z tests



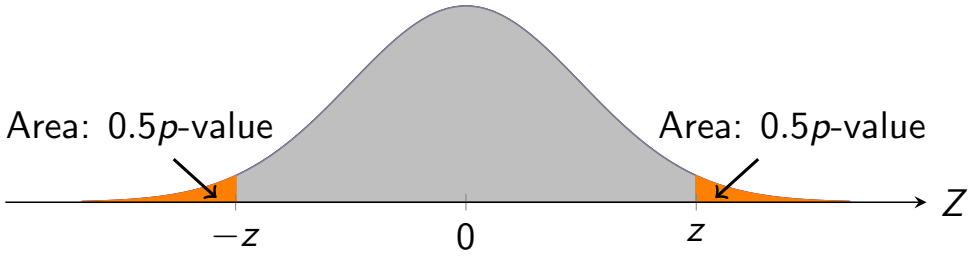
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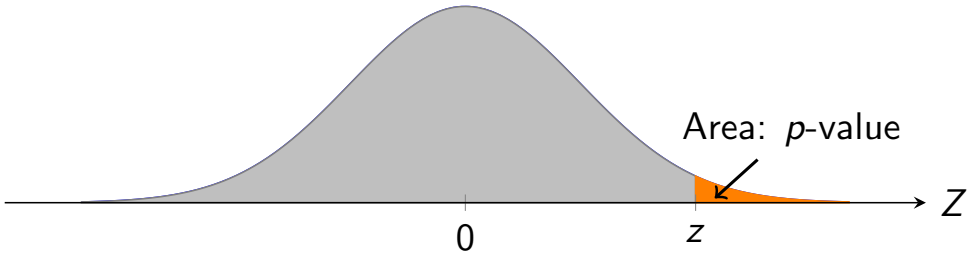
p-value: area in lower tail

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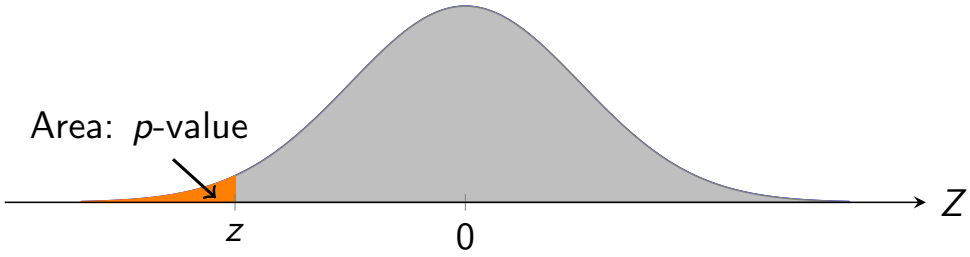
p-value: sum of area in both tails

p-value for z tests



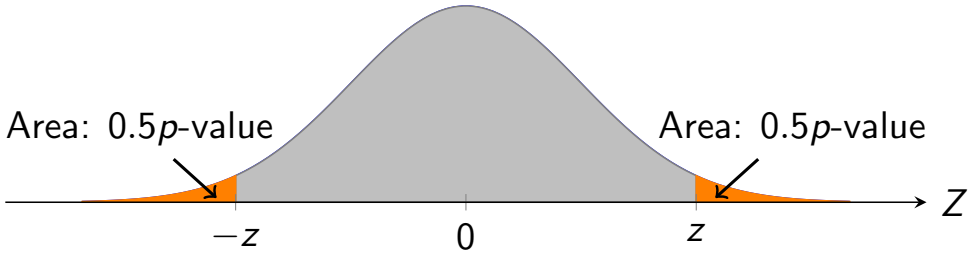
p-value: area in upper tail

$$p = 1 - \Phi(z) \quad (12)$$



p-value: area in lower tail

$$p = \Phi(z) \quad (13)$$



p-value: sum of area in both tails

$$p = 2(1 - \Phi(|z|)) \quad (14)$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (OS 5.21)

400 students were randomly sampled from a large university, and 289 said they did not get enough sleep. Conduct a hypothesis test to check whether this represents a statistically significant difference from 50%, and use a significance level of 0.01

Step 1. Parameter of interest:

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (OS 5.21)

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Notes:

$$p_0 = 0.50 \quad \text{(expected value)}$$

$$\hat{p} = \frac{289}{400} = .723 \quad \text{(observed value—from sample)}$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

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$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}}$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

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Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92))$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

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$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Step 7. Compare p-value to $\alpha = 0.01$:

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Step 7. Compare p-value to $\alpha = 0.01$:

$$0.0 \leq 0.01$$

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

Step 5. Calculate test statistic value:

$$z = \frac{.723 - .5}{\sqrt{.5(.5)/400}} = 8.92$$

Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Step 7. Compare p-value to $\alpha = 0.01$:

$$0.0 \leq 0.01$$

Step 8. Conclude:

Hypothesis testing using p-value approach

Example 4: Getting enough sleep (cont.)

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Step 6. Determine p-value (two-tailed test):

$$p\text{-value} = 2(1 - \Phi(8.92)) = 0.0$$

Step 7. Compare p-value to $\alpha = 0.01$:

$$0.0 \leq 0.01$$

Step 8. Conclude:
Using a significance level of 0.01, we reject H_0 since $0.0 < 0.01$.

Example 5: Equipment quality (p-value approach)

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A car manufacturer is considering switching to a new, higher quality piece of equipment that con- structs vehicle door hinges. They figure that they will save money in the long run if this new machine produces hinges that have flaws no more than 0.2% of the time. A random sample of 800 hinges produced by the new machine shows that 4 are flawed. At the 0.05 significance level, is there enough evidence to conclude that the new machine produces hinges with a flaw rate greater than 0.2%?

Example 5 (cont.)

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This is a one-sided test

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Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Example 5 (cont.)

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Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

$$\hat{p} = \frac{4}{800} = 0.005$$

Example 5 (cont.)

This is a one-sided test

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

$$\hat{p} = \frac{4}{800} = 0.005$$

$$SE = \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016$$

Example 5 (cont.)

This is a one-sided test

Hypotheses:

- $H_0 : p = 0.002$
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Test Statistic:

$$\begin{aligned} \hat{p} &= \frac{4}{800} = 0.005 \\ SE &= \sqrt{\frac{0.002(1-0.002)}{800}} \approx 0.0016 \\ z &= \frac{\hat{p} - 0.002}{SE} \approx 1.88 \end{aligned}$$

Example 5 (cont.)

This is a one-sided test

Hypotheses:

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Test Statistic:

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p-value: For $z = 1.88$, the p-value is:

$$p\text{-value} = 1 - \Phi(1.88) \quad [\texttt{1 - norm.cdf(1.88)}]$$

Example 5 (cont.)

This is a one-sided test

Hypotheses:

- $H_0 : p = 0.002$
- $H_a : p > 0.002$

Test Statistic:

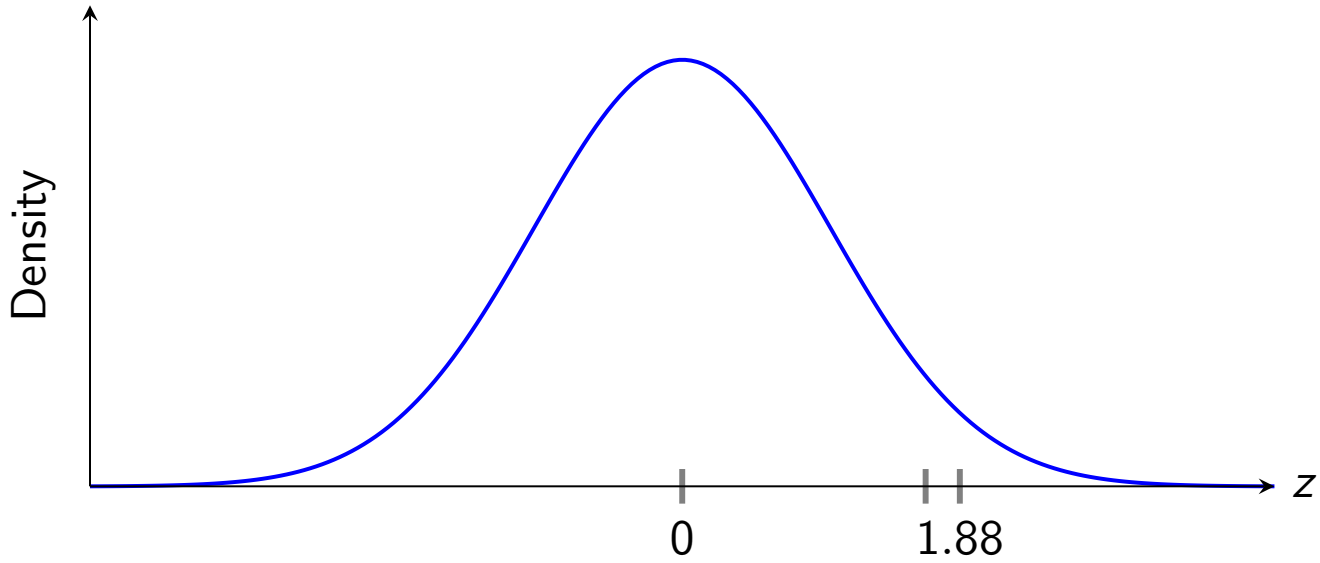
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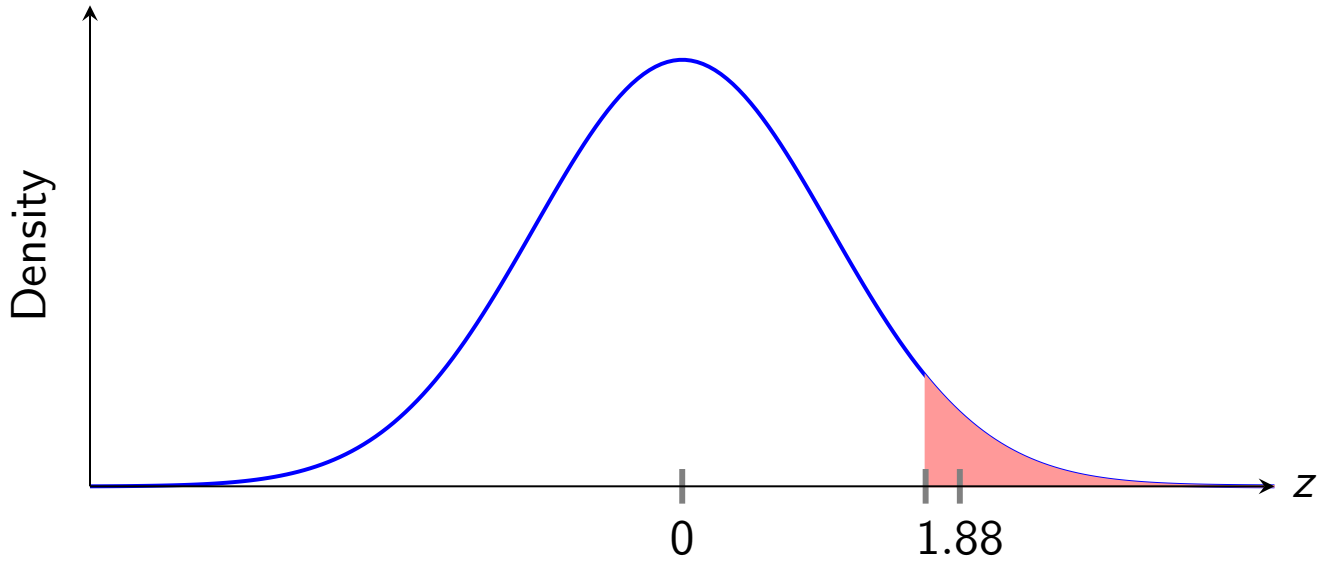
$$\begin{aligned} p\text{-value} &= 1 - \Phi(1.88) \quad [\texttt{1 - norm.cdf(1.88)}] \\ &\approx 0.0304 \end{aligned}$$

Example 5 (cont.)

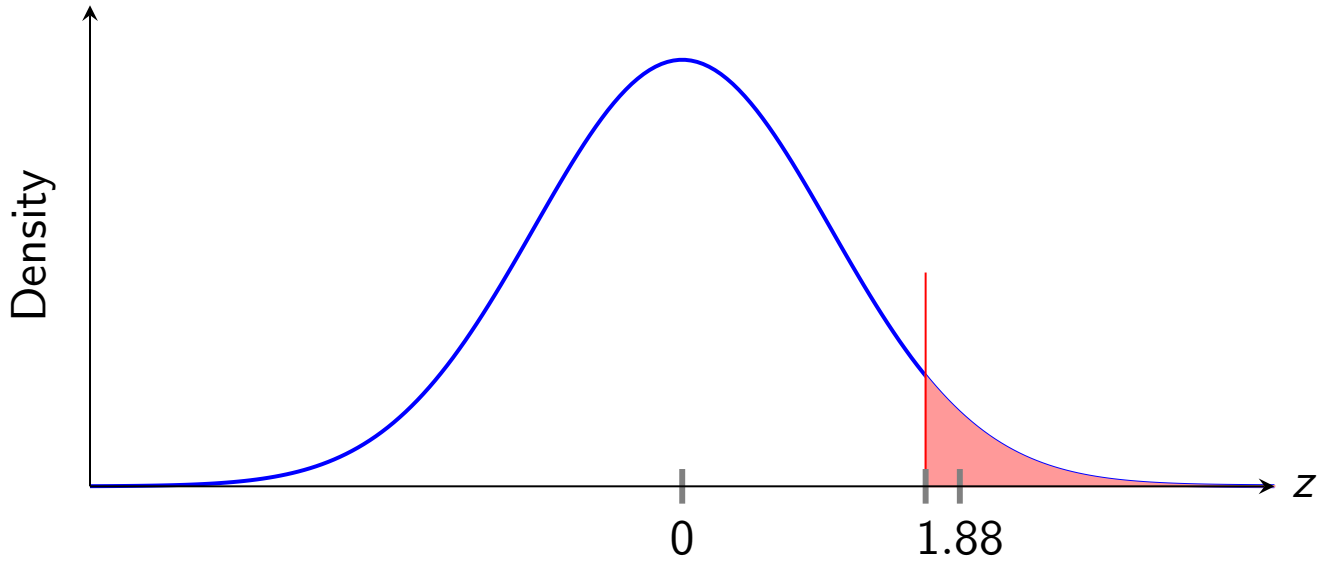
Example 5 (cont.)



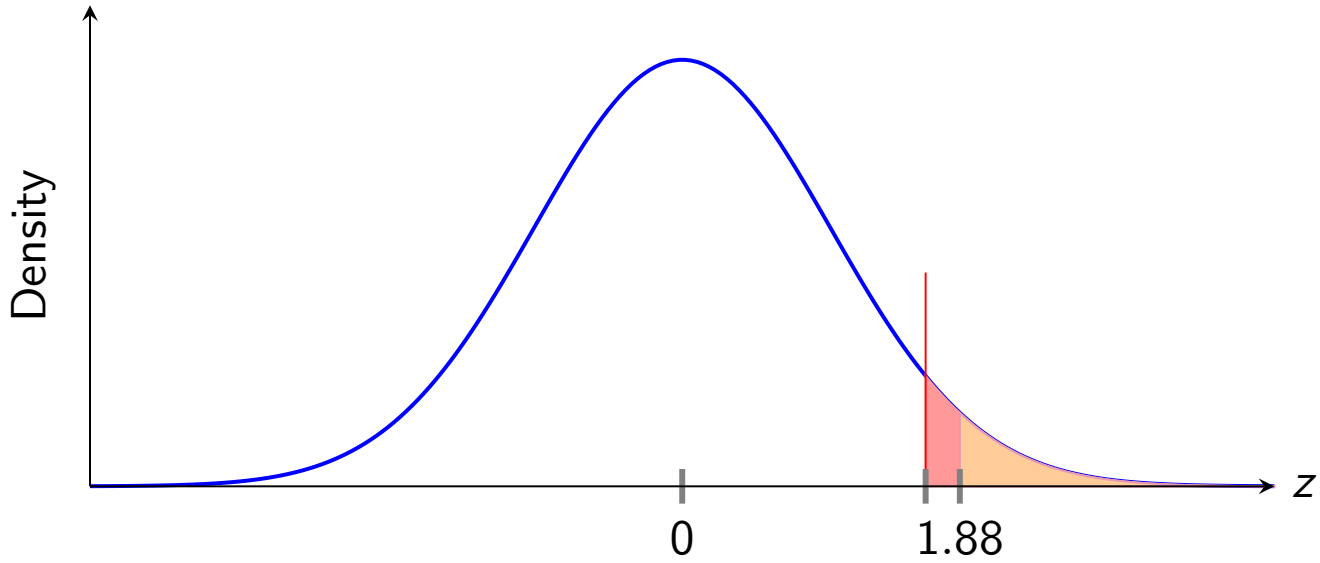
Example 5 (cont.)



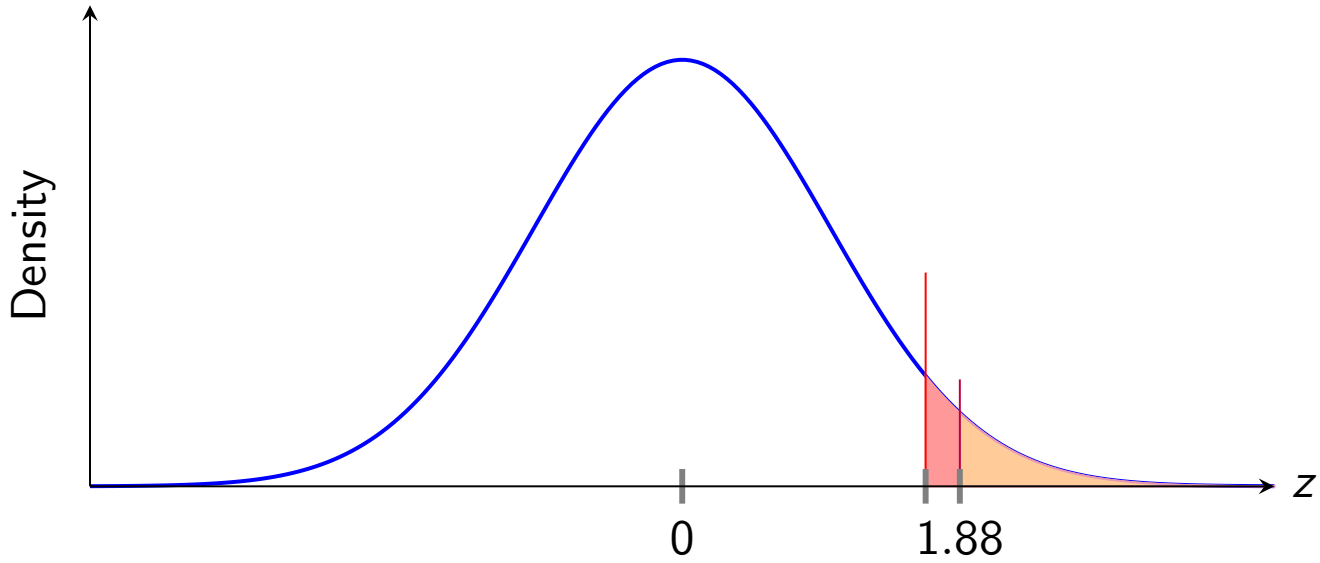
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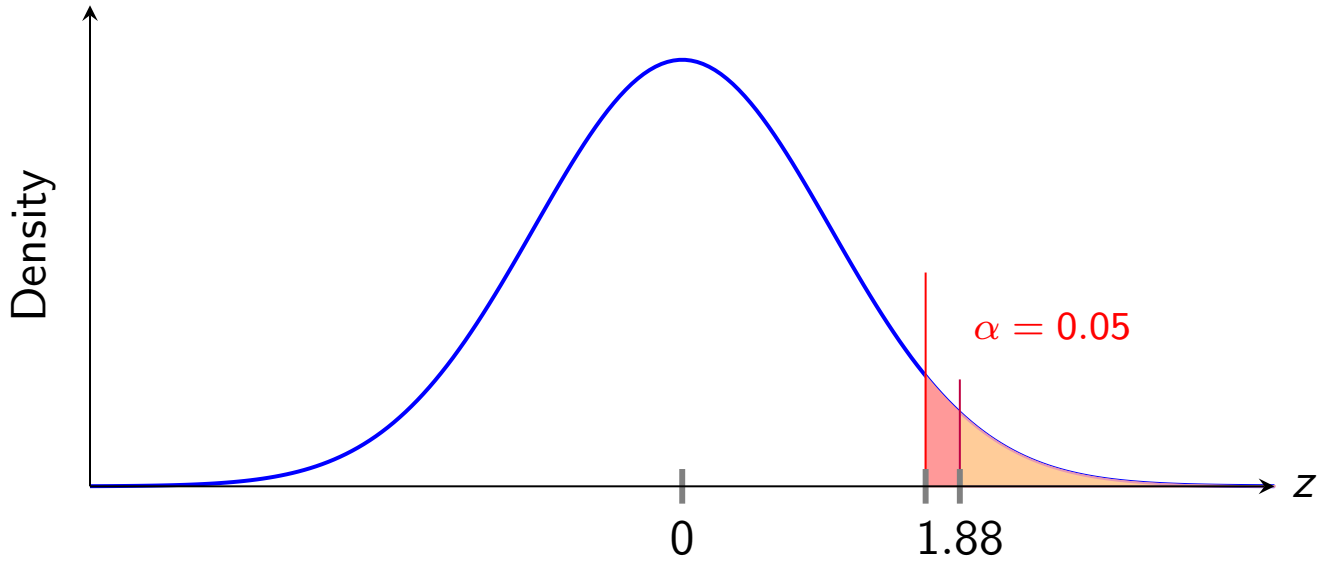
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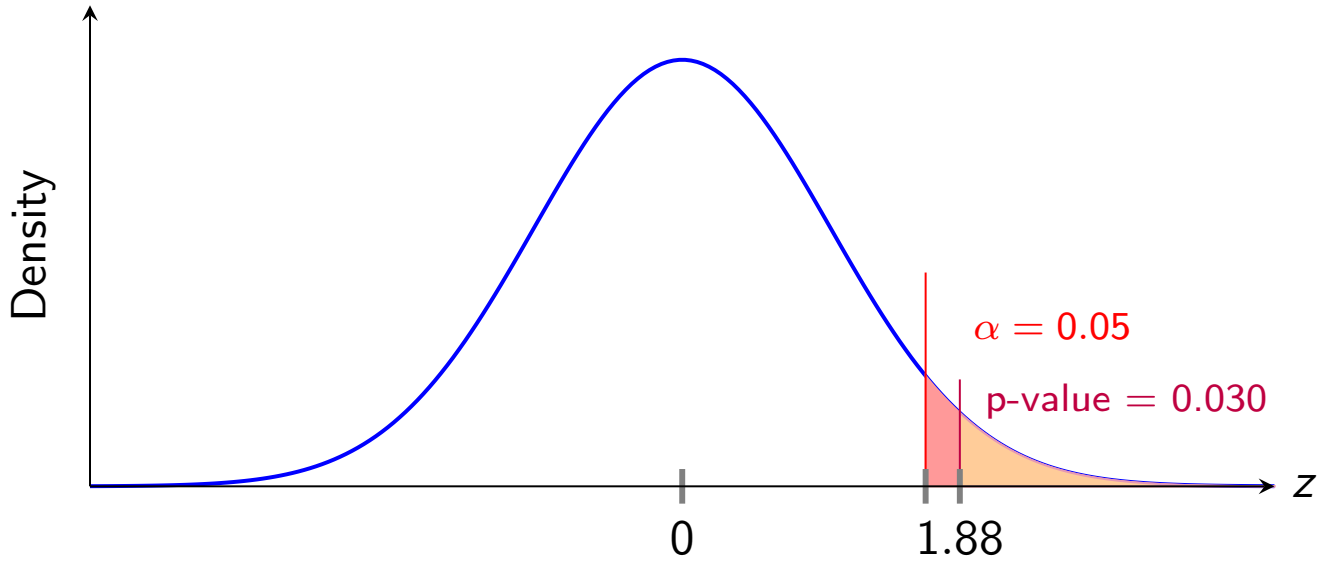
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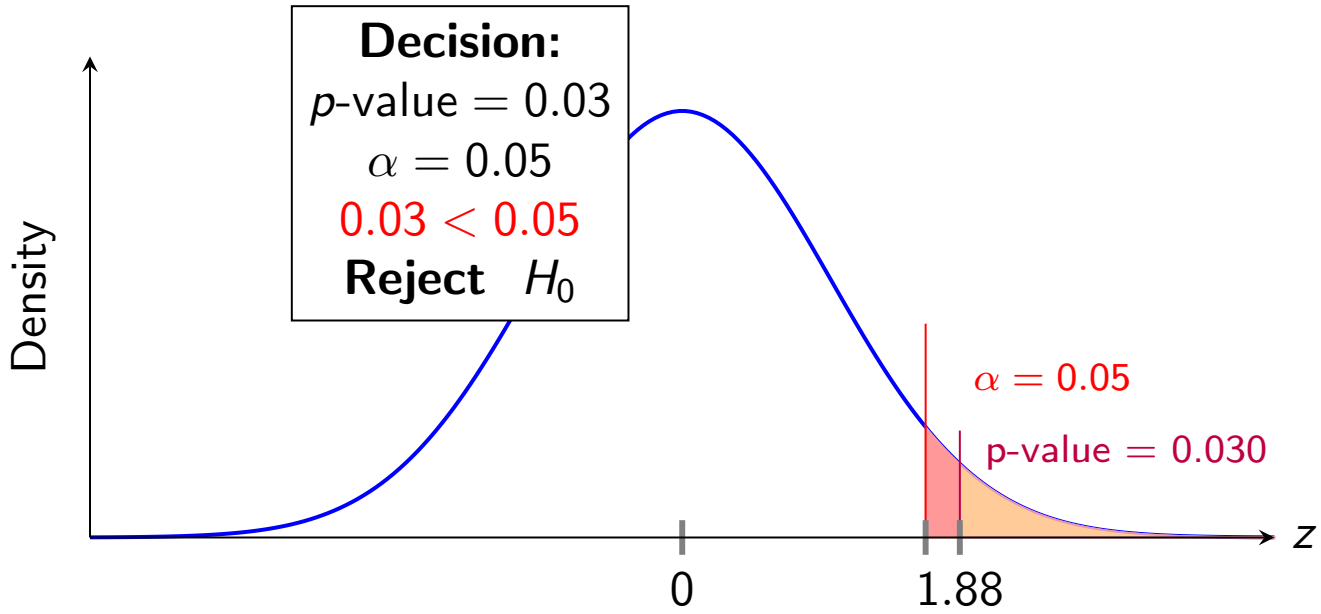
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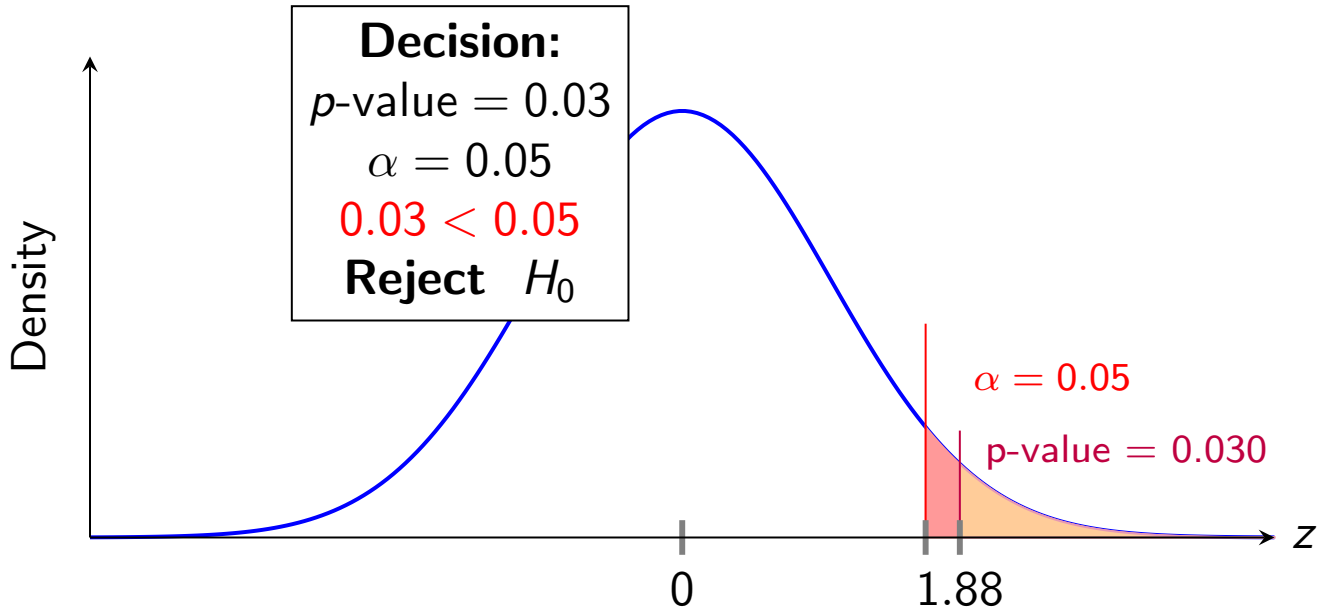
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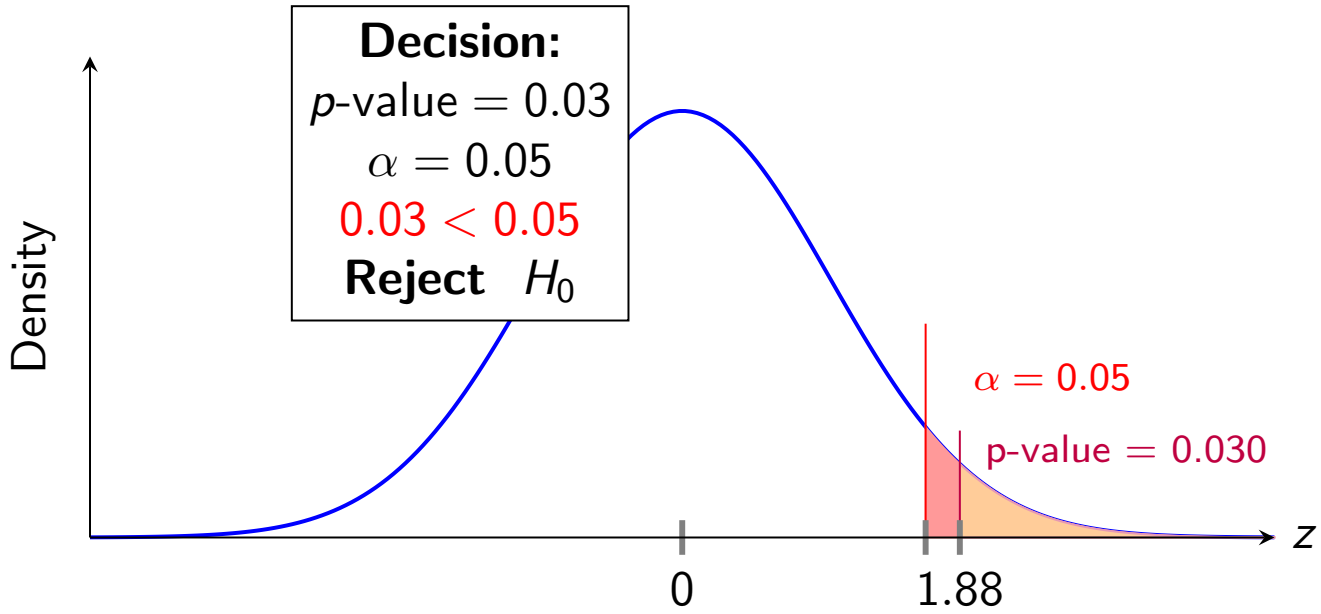
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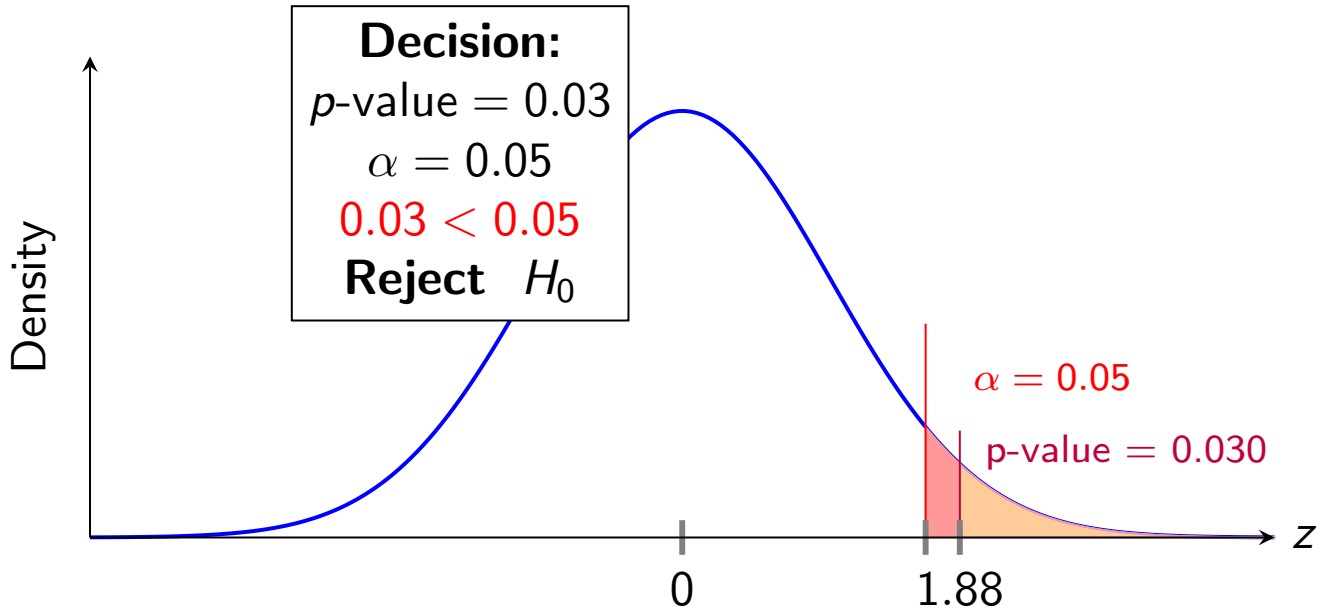


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Decision: Since the p-value $= 0.03 < \alpha = 0.05$, we **reject** H_0 at the 5% significance level.

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Conclusion: There is sufficient evidence to conclude that the new machine produces hinges with a flaw rate greater than 0.2%.

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